Entrepreneurship & Startup (MDM)
Assignment 2: Project Report

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MikroMile

1. Introduction

BuddyBasket (also referred to as MicroMile in the presentation) is a community-based micro-delivery system designed to solve the problem of urgent small-item needs. The core idea is simple: if someone is already going to a nearby store, they can bring small items for others in their community.

This report explains the working prototype developed, including tools used, system architecture, implementation details, and design decisions.

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2. Objective of the Working Prototype

The goal of the prototype was to validate the core concept of:

- Posting a “Going Out” notification
- Allowing users to request items
- Accepting requests
- Simulating delivery coordination

The focus was not on scaling but on demonstrating feasibility and user flow.

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3. Tools & Platforms Used

3.1 AI Tools Used

- Claude (Coding Agent):
  Used to generate structured full-stack code including frontend components, backend APIs, and database schema.
- Antigravity:
  Used for UI/UX ideation and refining frontend structure and layout.

These tools significantly reduced development time and helped in rapid prototyping.

3.2 Development Stack

Frontend:

- Next.js (React framework)
- Tailwind CSS for styling

Backend:

- Node.js
- Express.js

Database (Prototype):

- Local Storage (for quick testing and simulation)

Other:

- REST APIs for communication between frontend and backend

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4. System Architecture

The system follows a basic client-server architecture.

4.1 Frontend (Client)

- Built using Next.js
- Handles UI rendering and user interactions
- Sends API requests to backend

4.2 Backend (Server)

- Built using Express.js
- Handles business logic
- Manages trip creation, requests, and status updates

4.3 Data Layer

- Local storage used for storing:
    - users
    - trips
    - requests
- This was chosen for simplicity and faster testing

4.4 Architecture Flow

User → Frontend (Next.js) → API Calls → Backend (Express) → Local Storage → Response → UI Update

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5. Core Features Implemented

5.1 User Simulation

- Basic user identity system (name-based)
- No full authentication implemented (for prototype simplicity)

5.2 “Going Out” Feature

- Users can create a trip
- Inputs include:
    - destination (e.g., D-Mart)
    - time
    - notes
- Trip is displayed on dashboard

5.3 Request System

- Other users can request items
- Inputs:
    - item name
    - quantity
- Requests are linked to trips

5.4 Accept / Reject Flow

- Trip creator can:
  - accept requests
  - reject requests
- Status updates dynamically in UI

5.5 Status Tracking

Each request has states:

- pending
- accepted
- completed

5.6 UI Dashboard

- Shows all active trips
- Displays requests per trip
- Allows interaction through buttons

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6. Frontend Design Details

6.1 Framework Choice

Next.js was chosen because:

- Component-based architecture
- Fast rendering
- Easy routing

6.2 UI Design

- Clean, minimal interface
- Card-based layout
- Responsive design for mobile and desktop

6.3 Key Components

- TripCard

- RequestCard
- CreateTripForm
- RequestForm

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7. Backend Implementation

7.1 API Routes

Examples:

- POST /trips → create trip
- GET /trips → fetch trips
- POST /requests → create request
- PATCH /requests/:id → update status

7.2 Logic Handling

- Each trip contains multiple requests
- Requests are filtered based on trip ID
- State changes handled via API

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8. Database Design (Prototype)

Although local storage was used, the logical structure is:

User

- id
- name

Trip

- id
- userId
- destination
- time
- status

Request

- id
- tripId
- userId
- item
- quantity
- status

This structure can be easily migrated to MongoDB or SQL.

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9. Limitations of Current Prototype

- No real authentication system
- No real payment integration
- No GPS/location tracking
- No real-time notifications
- Data not persistent across sessions (local storage only)

These are expected in a prototype stage.

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10. Future Improvements

Technical

- Add authentication (Firebase/Auth)
- Use MongoDB for persistent database
- Integrate Google Maps API
- Add real-time notifications (Socket.io)
- Implement escrow payment system

Product

- Reward points system
- Ratings and reviews
- Community-based grouping
- Mobile app version

11. Conclusion

The BuddyBasket prototype successfully demonstrates the feasibility of a community-driven micro-delivery system. By leveraging existing foot traffic, the system eliminates the need for traditional delivery infrastructure.

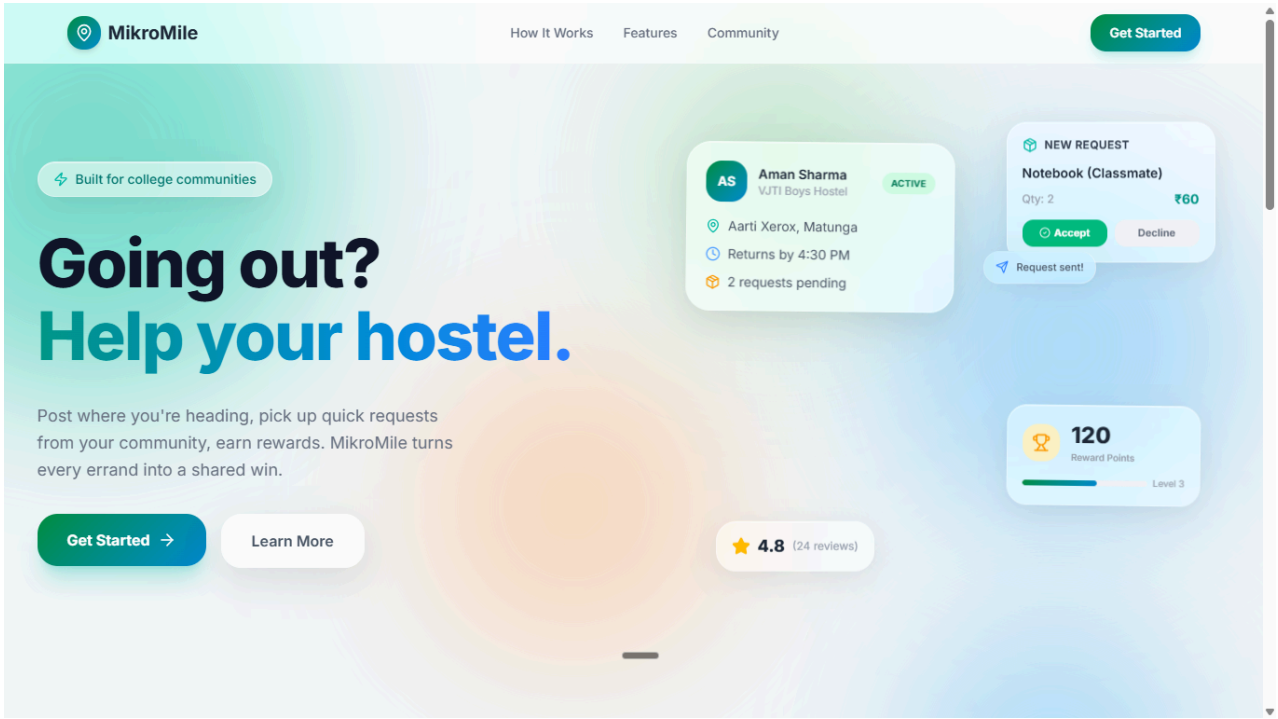
The prototype validates that:

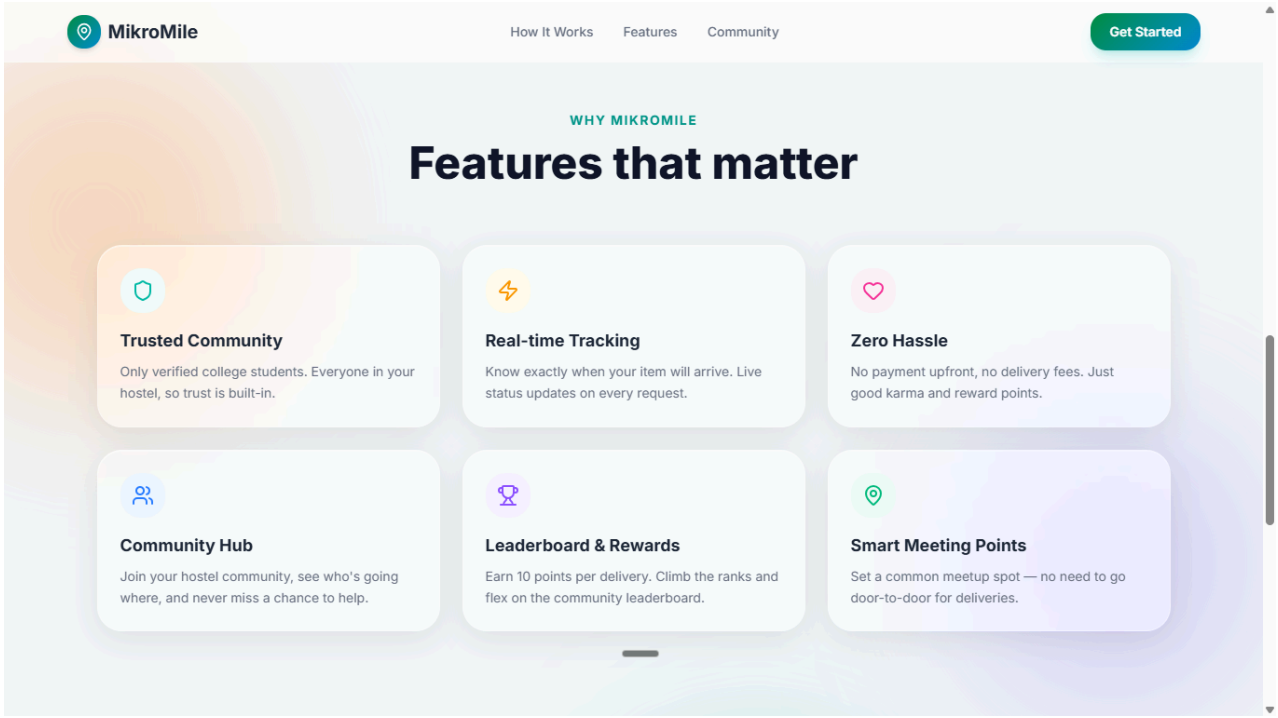
- Users can easily post trips
- Others can request items
- Requests can be managed efficiently

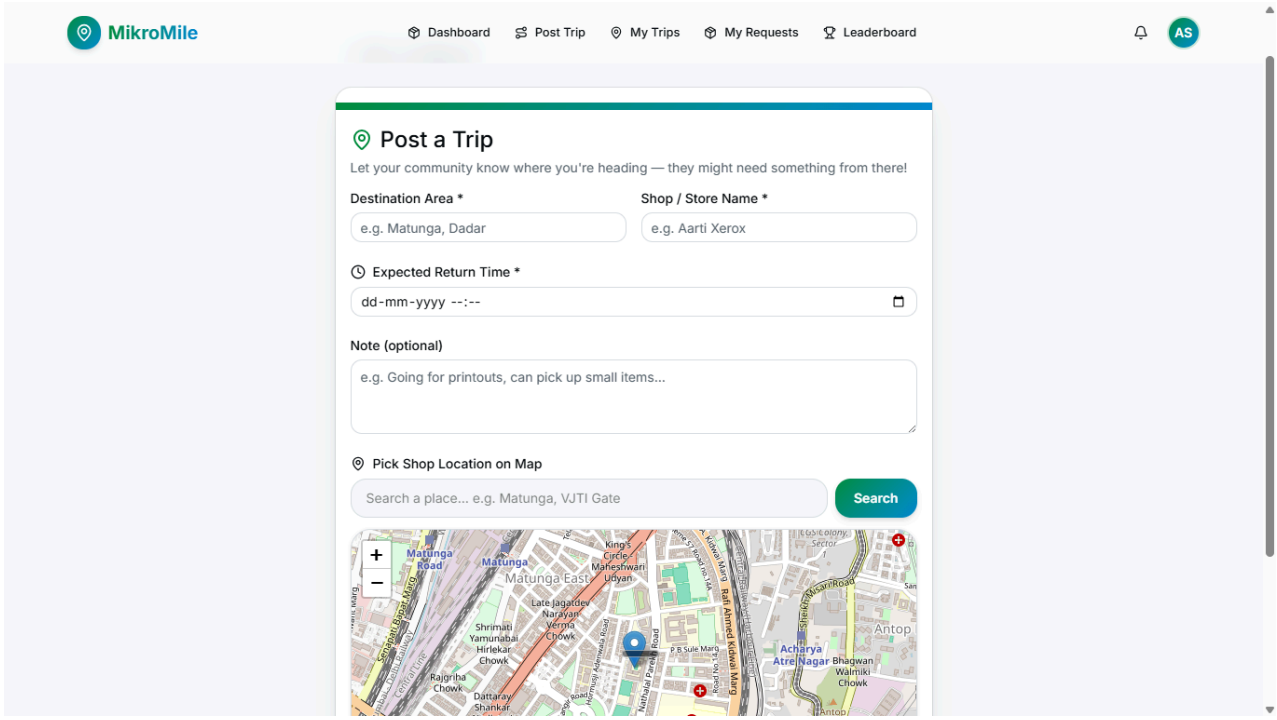
With further development, this idea has strong potential to scale within college campuses and residential communities.

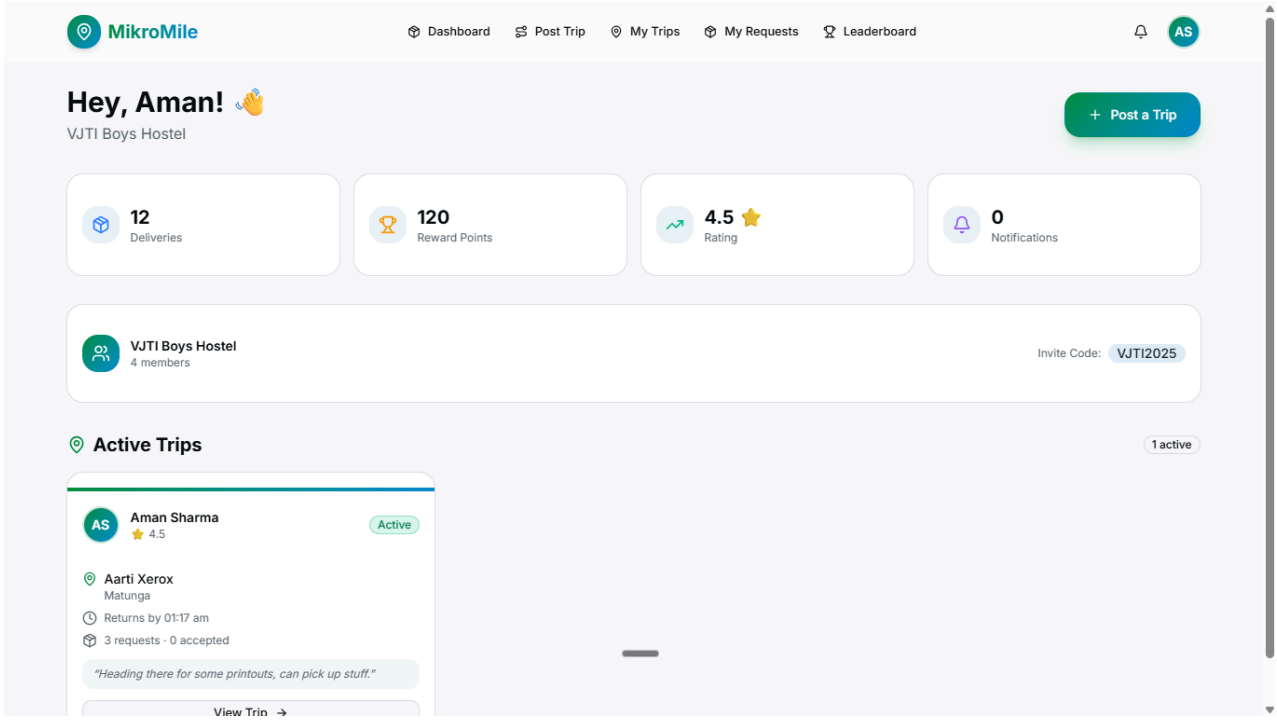
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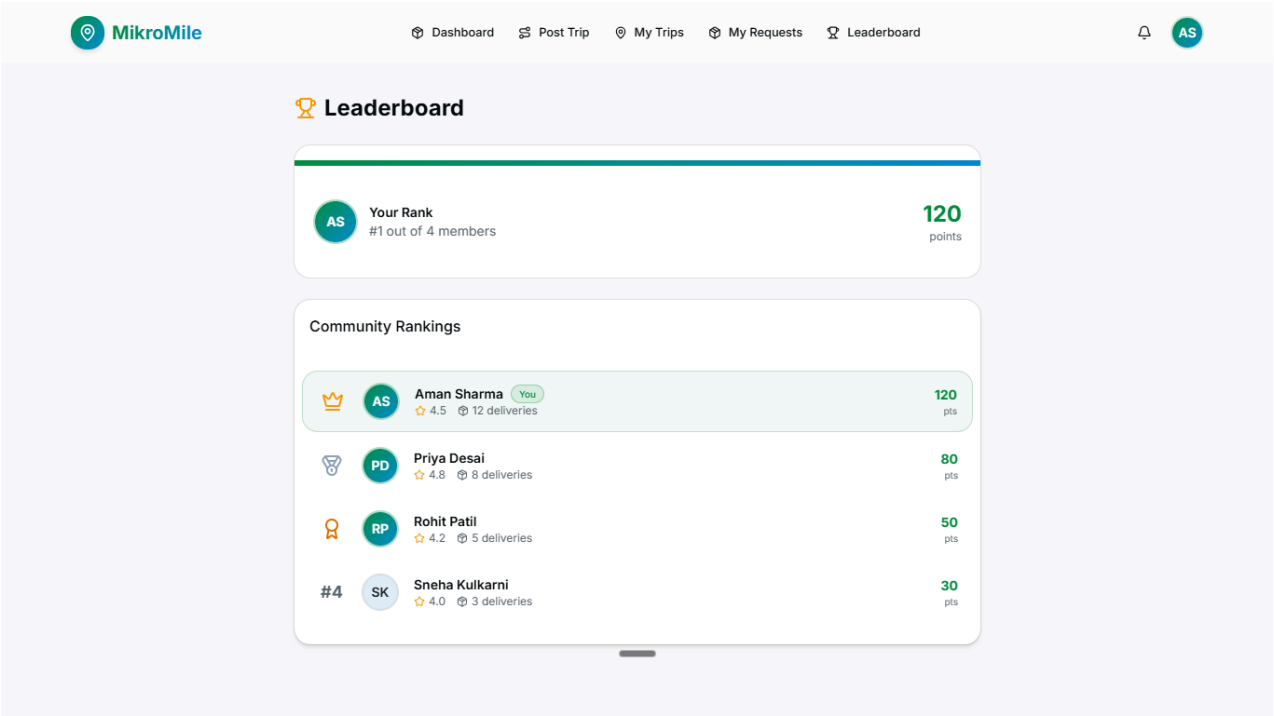
12. Working Demo and Screenshots




MikroMile
How It Works
Features
Community
Get Started
WHY MIKROMILE
Features that matter
Trusted Community
Only verified college students. Everyone in your hostel, so trust is built-in.
Real-time Tracking
Know exactly when your item will arrive. Live status updates on every request.
Zero Hassle
No payment upfront, no delivery fees. Just good karma and reward points.
Community Hub
Join your hostel community, see who's going where, and never miss a chance to help.
Leaderboard & Rewards
Earn 10 points per delivery. Climb the ranks and flex on the community leaderboard.
Smart Meeting Points
Set a common meetup spot — no need to go door-to-door for deliveries.


MikroMile
Dashboard
Post Trip
My Trips
My Requests
Leaderboard
AS
Post a Trip
Let your community know where you're heading — they might need something from there!
Destination Area *
e.g. Matunga, Dadar
Shop / Store Name *
e.g. Aarti Xerox
Expected Return Time *
dd-mm-yyyy --:--
Note (optional)
e.g. Going for printouts, can pick up small items...
Pick Shop Location on Map
Search a place... e.g. Matunga, VJTI Gate
Search
Matunga Road
Matunga
Matunga East
King's Circle Maheshwari Udyan
Late Jagatdev Narayan Verma Chowk
Shrimati Yamunabai Hirlekar Chowk
Rajgriha Chowk
Dattaray Shankar
Acharya Atre Nagar
Bhagwan Walmiki Chowk
Antop


MikroMile
Dashboard
Post Trip
My Trips
My Requests
Leaderboard
AS
Hey, Aman! 👋
VJTI Boys Hostel
+ Post a Trip
12
Deliveries
120
Reward Points
4.5
Rating
0
Notifications
VJTI Boys Hostel
4 members
Invite Code: VJTI2025
Active Trips
1 active
AS
Aman Sharma
4.5
Active
Aarti Xerox
Matunga
Returns by 01:17 am
3 requests · 0 accepted
"Heading there for some printouts, can pick up stuff."
View Trip →



https://drive.google.com/drive/folders/1MWVYO_U_BXP4q1c3Oz2wjvY-3dyiVpgF?usp=sharing